COURSE TITLE	**BLOCKCHAIN TECHNOLOGY & SECURITY**
COURSE CODE	**01CC0705**
COURSE CREDITS	**3**

Objective:

1 The course aims to provide students with a clear understanding of blockchain technology, including its core components such as blocks, chaining mechanisms, consensus algorithms, and peer-to-peer networks. It introduces fundamental cryptographic concepts like hashing, digital signatures, and public-private key encryption that ensure security and trust. The course also focuses on developing practical skills in writing and deploying smart contracts using Solidity and building decentralized applications (DApps) using Web3 technologies. Additionally, it enables students to analyze real-world blockchain applications across various domains and critically examine current challenges and emerging trends in blockchain technology.

Course Outcomes: After completion of this course, student will be able to:

1 Apply fundamental blockchain concepts and cryptographic techniques to explain the working of distributed ledger systems.

2 Implement smart contracts using Solidity to develop basic blockchain-based solutions.

3 Analyze the architecture and functionality of decentralized applications (DApps) using Web3 technologies.

4 Evaluate real-world blockchain applications and assess their effectiveness across domains such as finance, healthcare, and supply chain.

5 Design and develop secure and efficient blockchain-based applications while addressing challenges like scalability and interoperability.

Pre-requisite of course: Basic Programming Knowledge, Understanding of Data Structures, Fundamentals of Cryptography

Teaching and Examination Scheme

Theory Hours	Tutorial Hours	Practical Hours	ESE	IA	CSE	Viva	Term Work
2	0	2	50	30	20	25	25

Contents : Unit	Topics	Contact Hours
1	**Introduction to Blockchain** What is Blockchain, Blockchain Technology Mechanisms & Networks, Blockchain Origins,, Objective of Blockchain, Blockchain Challenges, Transactions And Blocks,, P2P Systems, Keys As Identity, Digital Signatures,, Hashing and public key cryptosystems, Private vs. Public Blockchain	6

Contents : Unit	Topics	Contact Hours
2	**Models of blockchain** Introduction, Changing of Blocks, Hashing, Merkle-Tree, Consensus, Mining and Finalizing Blocks, Currency aka tokens, Security on blockchain, Data storage on blockchain, Wallets, Coding on blockchain: Smart Contracts, Peer-to-peer Network, Types of blockchain nodes, Risk associated with blockchain solutions, Life cycle of blockchain transaction.	6
3	**Mechanics Of Blockchain** Bitcoin, Wallet, Blocks, Merkley Tree, Bitcoin transactions, Transaction verifiability, Anonymity, Forks, Double spending, Mathematical analysis of properties of Bitcoin, Bitcoin scripts, Applications of Bitcoin scripts, Bitcoin blocks, The Bitcoin network, Limitations and improvements, How to store and use Bitcoins, Simple local storage, Hot and cold storage, Splitting and sharing keys, Online wallets and exchanges,, Payment services, Transaction fees, Currency exchange markets.	10
4	**Alternative Coins And Recent Trends** Ethereum, Ethereum Virtual Machine (EVM),, Wallets for Ethereum, Solidity, Smart Contracts, Some attacks on smart contracts, Zero Knowledge proofs and protocols in Blockchain, Succinct non interactive argument for Knowledge (SNARK),, Pairing on Elliptic curves, Zcash.	6
5	**WEB 3.0** From 1.0 to 3.0, Decentralization, Encryption, Tokenization, Social Engineering and new form of attack, Data security and Reliability,, Identity and anonymity.	4
Total Hours		**32**

Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
1	**Practical 1** Set Up a Private Ethereum Blockchain Network Objective: Learn how to configure and initialize a private Ethereum network. Tool: Geth (Go Ethereum) – Open-source.	2
2	**Practical 2** Create and Deploy a Simple Smart Contract Objective: Write a basic smart contract (e.g., "Hello Blockchain") in Solidity and deploy it. Tools: Remix IDE – Browser-based open-source Solidity IDE.	2
3	**Practical 3** Connect to Blockchain Using Web3.js Objective: Interact with deployed smart contracts using JavaScript and Web3.js. Tools: Web3.js with Node.js.	2

Suggested List of Experiments:

Contents : Unit	Topics	Contact Hours
4	**Practical 4** Create a Decentralized Voting DApp Objective: Build a DApp (frontend + smart contract) for a simple voting system. Tools: Remix, Web3.js, Ganache (for testing), Metamask.	2
5	**Practical 5** Simulate Blockchain Consensus Mechanisms Objective: Demonstrate Proof of Work (PoW) or Proof of Stake (PoS) mechanisms through code simulation. Tool: Python or JavaScript (custom implementation).	2
6	**Practical 6** Perform Transactions on the Ethereum Blockchain Objective: Use Geth or Remix to send test Ether between accounts on a private blockchain. Tool: Geth / Remix (no real Ether required).	2
7	**Practical 7** Use Hyperledger Fabric to Deploy a Blockchain Network Objective: Set up a Hyperledger Fabric network and deploy a simple chaincode. Tool: Hyperledger Fabric – Open-source blockchain platform.	2
8	**Practical 8** Explore NFT Creation and Deployment on Testnet Objective: Mint and deploy a Non-Fungible Token (NFT) using ERC-721 standard on a test network. Tools: Remix, Metamask, OpenZeppelin libraries, Rinkeby or Sepolia testnet.	2
9	**Practical 9** Record Student Records Using Blockchain Objective: Build a DApp to store and retrieve immutable student data using smart contracts. Tools: Solidity, Remix, Web3.js, Ganache.	2
10	**Practical 10** Blockchain-Based Supply Chain Tracking Objective: Simulate a blockchain system to track product movement across a supply chain. Tools: Remix for contracts, basic web frontend with Web3.js, Ganache.	2
Total Hours		**20**

Textbook :

1 Blockchain Security: Building Confidence in a Trustless World, Kevin D. Werbach and Ittay Eyal, -, -

References:

1 Blockchain Basics: A Comprehensive Introduction to Cryptocurrency, Blockchain, and Blockchain Security, Blockchain Basics: A Comprehensive Introduction to Cryptocurrency, Blockchain, and Blockchain Security, Michael Casey and Paul Vigna, -, 2020

References:

2 Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications, Mastering Blockchain: Unlocking the Power of Cryptocurrencies, Smart Contracts, and Decentralized Applications, Andreas M. Antonopoulos, Shroff/O'Reilly, -

Suggested Theory Distribution:

The suggested theory distribution as per Bloom's taxonomy is as follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery

Remember / Knowledge	Understand	Apply	Analyze	Evaluate	Higher order Thinking / Creative
0.00	7.00	30.00	33.00	20.00	10.00

Instructional Method:

1 The course delivery method will depend upon the requirement of content and need of students. The teacher in addition to conventional teaching method by black board, may also use any of tools such as demonstration, role play, Quiz, brainstorming, MOOCs etc.

2 The internal evaluation will be done on the basis of continuous evaluation of students in the laboratory and class-room.

3 Practical examination will be conducted at the end of semester for evaluation of performance of students in laboratory

4 Students will use supplementary resources such as online videos, NPTEL videos, e-courses, Virtual Laboratory.

Supplementary Resources:

1 "Cisco Networking Academy" – Packet Tracer Course

2 "Computer Networking" A Top-Down Approach" by Kurose and Ross

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